

Fitness Business Data Analysis

Data Analyst Capstone — Final Explanation Report

Dataset period: December 2020

Tools: Python / Pandas, Excel, Power BI

Objective: Analyze fitness-session performance, identify useful patterns, and present stakeholder-ready insights through a BI dashboard.

1. Project Objective

The project converts the supplied fitness-session data into a repeatable analytics workflow: data quality checks in Python, grouped analysis in Pandas/Excel, and a Power BI dashboard for interactive stakeholder exploration.

2. Dataset & Data Quality

Metric	Result
Rows / columns	30 / 11
Missing values	0
Duplicate rows	0
Date range	2020-12-01 to 2020-12-31
Intensity levels	Medium, High, Low

The supplied dataset contains 30 records and 10 columns. The quality check found no missing values and no duplicate rows.

3. KPI Summary

KPI	Value
Total Sessions	30
Total Calories	9,218.7
Average Calories / Session	307.29
Average Calories / Minute	5.55
Average Duration	55.5 min
Average Pulse	102.8
Maximum Maxpulse	175

4. Analysis Approach

Step 1 — Python: Load the Excel data with Pandas, convert Date to a date type, inspect dimensions/types, test missing values and duplicates, and validate the Cal_per_Min calculated field.

Step 2 — EDA: Calculate headline KPIs and grouped summaries by Intensity, Weekday, and Week.

Step 3 — Excel: Preserve the source data and produce analysis-ready grouped sheets for review.

Step 4 — Power BI: Load the dataset, create reusable DAX measures, and build an interactive stakeholder dashboard.

5. Dashboard Design

The dashboard follows the supplied interactive prototype: a dark header, filter controls for Intensity and Weekday, four KPI cards, a daily calorie trend, average calories by intensity, and a session-detail table. The prototype explicitly defines these filters, KPI labels, chart titles, and table fields. ■filecite■turn4file0■L35-L56■

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The course brief requires the project to clean/explore the dataset in Python, analyze it using SQL or Pandas, visualize it using Excel and a BI tool, and compile a final report with charts, findings, and recommendations.

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6. Key Findings

- The dataset contains 30 logged sessions across December 2020.
- Total recorded calories burned are 9,218.7, averaging 307.29 calories per session.
- Average calories per minute are 5.55, while the average session duration is 55.5 minutes.
- Medium intensity is the dominant category in the supplied records; the dashboard therefore needs an intensity filter so stakeholders can isolate Low, Medium, or High sessions.
- The maximum recorded Maxpulse is 175. This is a descriptive statistic from the dataset, not a clinical interpretation.

7. Recommendations

- Use the Intensity and Weekday slicers during stakeholder reviews to compare different session segments rather than relying only on overall averages.
- Track daily calories alongside intensity to identify unusually high or low activity days.
- Use Cal_per_Min as an efficiency metric alongside total calories, because total calories can increase simply with longer sessions.
- Keep the dashboard focused on the supplied business metrics and avoid unsupported causal claims from this small dataset.

8. Conclusion

The completed workflow satisfies the requested capstone structure: Python EDA, Pandas-based analysis, Excel summaries, Power BI dashboard specification, and a stakeholder-oriented report. The package is organized so the code, report, source data, and Power BI instructions can be submitted together.